Introduction to Mathematics and Optimisation

Exercises Week 6

Kirsten Pleskot (202507620)
20. 01. 2026

1 Exercise 5.22

A circle with center (a, b) and radius r is given by the equation

$$(x-a)^2 + (y-b)^2 = r^2.$$

(1) Explain how (5.14) can be rewritten to the equation

$$2ax + 2by + c = x^2 + y^2,$$

where $c = r^2 - a^2 - b^2$.

(2) Explain how fitting a circle to the points $(x_1, y_1), \ldots, (x_n, y_n)$ in the least squares context using (5.15) leads to the system

$$\begin{pmatrix} 2x_1 & 2y_1 & 1 \\ 2x_2 & 2y_2 & 1 \\ \vdots & \vdots & \vdots \\ 2x_n & 2y_n & 1 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} x_1^2 + y_1^2 \\ x_2^2 + y_2^2 \\ \vdots \\ x_n^2 + y_n^2 \end{pmatrix},$$

of linear equations.

(3) Compute the best circle through the points

$$(0,2), \quad (0,3), \quad (2,0) \quad \text{and} \quad (3,1)$$

1.1 Rewriting the Circle Equation

$$c = r^2 - a^2 - b^2 \tag{1}$$

$$(x-a)^2 + (y-b)^2 i = r^2 \tag{2}$$

$$x^2 - 2ax + a^2 + y^2 - 2by + b^2 = r^2 \tag{3}$$

$$2ax + 2by + r^2 - a^2 - b^2 = x^2 + y^2 \tag{4}$$

$$2ax + 2by + c = x^2 + y^2 \text{ by substituting (1)} \tag{5}$$

1.2 Fitting a circle using least squares method

Given point $\{(x_1, y_1), (x_2, y_2), \dots, (x_i, y_i)\}$, the perfect circle would satisfy these equations:

- $2ax_1 + 2by_1 + c = x_1^2 + y_1^2$
- $2ax_2 + 2by_2 + c = x_2^2 + y_2^2$
- $\dots$
- $2ax_i + 2by_i + c = x_i^2 + y_i^2$

The formulas can be written in a form $Ax = b$, where

$$x = \begin{pmatrix} a \\ b \\ c \end{pmatrix}, \quad A = \begin{pmatrix} 2x_1 & 2y_1 & 1 \\ 2x_2 & 2y_2 & 1 \\ \vdots & \vdots & \vdots \\ 2x_i & 2y_i & 1 \end{pmatrix}, \quad b = \begin{pmatrix} x_1^2 + y_1^2 \\ x_2^2 + y_2^2 \\ \vdots \\ x_i^2 + y_i^2 \end{pmatrix}$$

Giving us the system:

$$\begin{pmatrix} 2x_1 & 2y_1 & 1 \\ 2x_2 & 2y_2 & 1 \\ \vdots & \vdots & \vdots \\ 2x_i & 2y_i & 1 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} x_1^2 + y_1^2 \\ x_2^2 + y_2^2 \\ \vdots \\ x_i^2 + y_i^2 \end{pmatrix}$$

This system in this form can be used to find the least squares solution for a, b and c.

1.3 Computing the best circle through given points

Given points: $(0, 2), (0, 3), (2, 0), (3, 1)$

$$A = \begin{pmatrix} 0 & 4 & 1 \\ 0 & 6 & 1 \\ 4 & 0 & 1 \\ 6 & 2 & 1 \end{pmatrix}, \quad b = \begin{pmatrix} 4 \\ 9 \\ 4 \\ 10 \end{pmatrix}$$

To find the least squares solution, we solve the normal equations:

$$A^\top A x = A^\top b$$

$$A^\top A = \begin{pmatrix} 52 & 12 & 10 \\ 12 & 56 & 12 \\ 10 & 12 & 4 \end{pmatrix} \tag{6}$$

$$A^\top b = \begin{pmatrix} 76 \\ 90 \\ 27 \end{pmatrix} \tag{7}$$

$$\begin{pmatrix} 52 & 12 & 10 \\ 12 & 56 & 12 \\ 10 & 12 & 4 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 76 \\ 90 \\ 27 \end{pmatrix} \tag{8}$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} \frac{83}{54} \\ \frac{11}{6} \\ -\frac{70}{27} \end{pmatrix} \tag{9}$$

The best fitting circle has center $\left(\frac{83}{54}, \frac{11}{6}\right)$ and radius calculated as follows:

$$r^2 = a^2 + b^2 + c \tag{10}$$

$$r = \frac{83}{54}^2 + \frac{11}{6}^2 - \frac{70}{27} \tag{11}$$

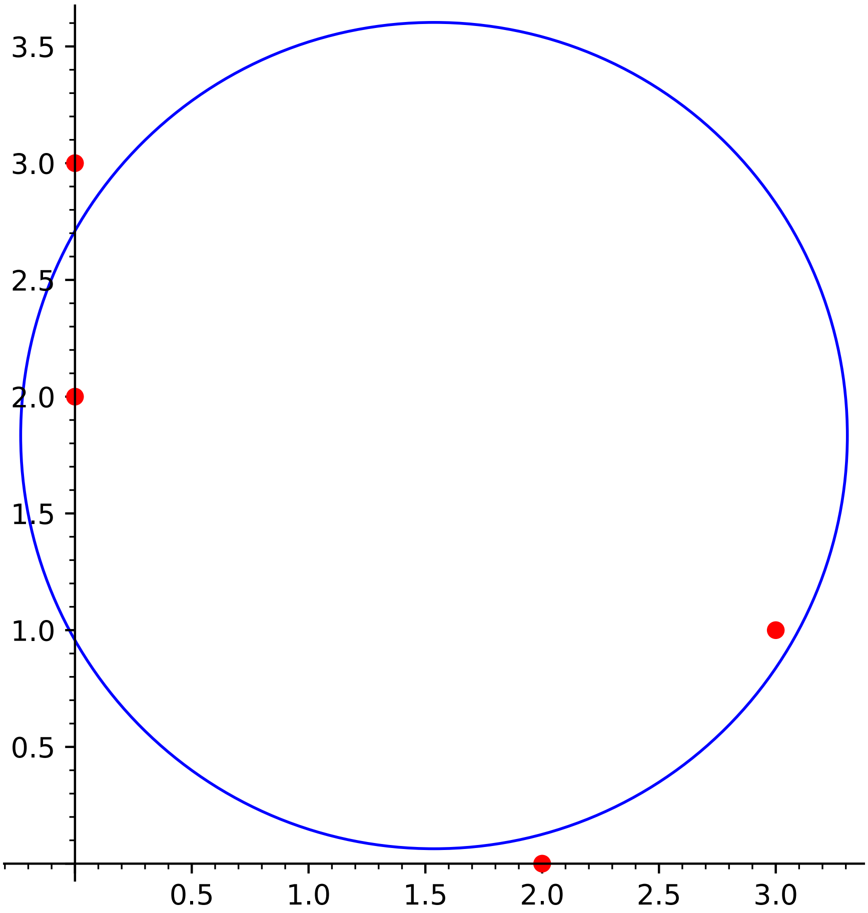
$$r = \frac{1}{27}\sqrt{\frac{4575}{2}} \tag{12}$$

$$r^2 = \frac{4565}{1458} \tag{13}$$

Giving us the final result: $\boxed{\left(x - \frac{83}{54}\right)^2 + \left(y - \frac{11}{6}\right)^2 = \frac{4565}{1458}}$

1.4 Plotting

The following values produce this plot in sage:



The exercise was solved using this sage script:

```
points = [(0, 2), (0, 3), (2, 0), (3, 1)]

A_rows = []
b_entries = []

# Use QQ for exact rational arithmetic
for x, y in points:
    A_rows.append([2*x, 2*y, 1])
    b_entries.append(x^2 + y^2)

A = matrix(QQ, A_rows)
b = vector(QQ, b_entries)

print("Matrix A:")
print(A)
```

```
print("Vector b:")
print(b)
print()

# Solve the equations: (A^T * A) * v = (A^T * b)
At = A.transpose()
AtA = At * A
Atb = At * b
print("AtA:")
print(AtA)
print("Atb:")
print(Atb)

v = AtA.solve_right(Atb)

print(f"v = {v}")

a = v[0]
b = v[1]
c = v[2]

center = (a, b)
r_squared = c + a^2 + b^2
r = sqrt(r_squared)

print(f"Center (a, b) = ({a}, {b})")
print(f"Radius r = {r}")
print(f"Equation: (x - {a})^2 + (y - {b})^2 = {r_squared}")

plot_points = list_plot(points,
                        color='red',
                        size=40,
                        zorder=2)

plot_circle = circle(center, r,
                     color='blue',
                     zorder=1)

final_plot = plot_points + plot_circle

final_plot.save('best_fit_circle.svg')
```